🔑 **Answer Key — Teacher Copy · Lesson 2-6**

MSTAR PRACTICE WORKSHEET

Lesson 2-6: Divide Multi-Digit Numbers Using an Algorithm

Grade 6 Mathematics · Standard 6.NOS.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is 936 ÷ 12?

A. 68

B. 76

C. 78

D. 87

12 × 78 = 936. You can verify: 12 × 70 = 840, 12 × 8 = 96, 840 + 96 = 936.

Question 2 SELECTED RESPONSE — CORRECT: C

What is 2,485 ÷ 5?

A. 487

B. 496

C. 497

D. 507

5 × 497 = 2,485. Running the cycle: 24 ÷ 5 → 4 with 4 left, 48 ÷ 5 → 9 with 3 left, 35 ÷ 5 → 7 with 0 left, so the quotient is 497 and the remainder is 0.

Question 3 SELECTED RESPONSE — CORRECT: B

What is 756 ÷ 9?

A 83

B 84

C 86

D 94

9 × 84 = 756. Running the cycle: 75 ÷ 9 → 8 with 3 left, then 36 ÷ 9 → 4 with 0 left, so the quotient is 84 and the remainder is 0.

Question 4 SELECTED RESPONSE — CORRECT: C

What is 1,125 ÷ 9?

A 115

B 124

C 125

D 135

9 × 125 = 1,125. Running the cycle: 11 ÷ 9 → 1 with 2 left, 22 ÷ 9 → 2 with 4 left, 45 ÷ 9 → 5 with 0 left, so the quotient is 125.

Question 5 SELECTED RESPONSE — CORRECT: C

A warehouse has 2,184 items to pack into boxes of 14. How many boxes are needed?

A 146 boxes

B 155 boxes

C 156 boxes

D 166 boxes

2,184 ÷ 14 = 156. Check: 14 × 156 = 14 × 150 + 14 × 6 = 2,100 + 84 = 2,184.

Question 6 SELECTED RESPONSE — CORRECT: C

What is 4,896 ÷ 12?

A 48

B 380

C 408

D 480

12 × 408 = 4,896.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Use the long division algorithm to find 936 ÷ 4.

A 209

B 232

C 234

D 243

9 ÷ 4 = 2 remainder 1; bring down 3 to make 13. 13 ÷ 4 = 3 remainder 1; bring down 6 to make 16. 16 ÷ 4 = 4. The quotient is 234.

- **A:** A digit was skipped. After 9 divided by 4 leaves remainder 1, bring the 3 down to make 13, which gives the tens digit.
- **B:** The last step is 16 divided by 4, which is 4. Check the final digit after bringing down the 6.
- **D:** The tens and ones digits are swapped. Work left to right: 2, then 3, then 4.

Part B — correct answer: B

In the D-M-S-B cycle, what does the SUBTRACT step tell you?

A How many times the divisor fits into the whole dividend.

B How much of the current value is left over before the next digit is brought down.

C Whether the quotient will be even.

D The final remainder of the entire problem.

Subtracting the product from the current partial dividend leaves the remainder at that place, which the next digit is appended to.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: The Dropped Zero

> Andre computed 618 ÷ 3. He wrote: '6 ÷ 3 = 2, then 1 ÷ 3 does not go so I skip it, then 8 ÷ 3 = 2 remainder 2. My answer is 22 remainder 2.'

Explain Andre's misconception and give the correct quotient.

Model answer: 618 ÷ 3 = 206. When 1 ÷ 3 does not go, the quotient gets a 0 in the tens place and the 1 stays to make 18 with the next digit; 18 ÷ 3 = 6. Skipping the place entirely loses the value of that digit.

2 points	Student explains that when a digit is too small the quotient gets a 0 in that place and the digit is carried to the next step (1 becomes 18), and gives 206.
1 point	Student gives 206 without explaining the placeholder zero.
0 points	Student agrees with Andre or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

The station must load 3,360 ration packs equally into 14 storage pods. Solve it with long division. Name each step — divide, multiply, subtract, bring down — and check your answer by multiplying.

Model answer: Long division on 3,360 ÷ 14 runs the cycle three times: 33 ÷ 14 → 2 (2 × 14 = 28, leaving 5), bring down the 6 → 56 ÷ 14 → 4 (4 × 14 = 56, leaving 0), bring down the 0 → 0 ÷ 14 → 0, so a 0 must be written in the ones place. The quotient is 240 ration packs per pod, and 14 × 240 = 3,360 proves none were left over.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
